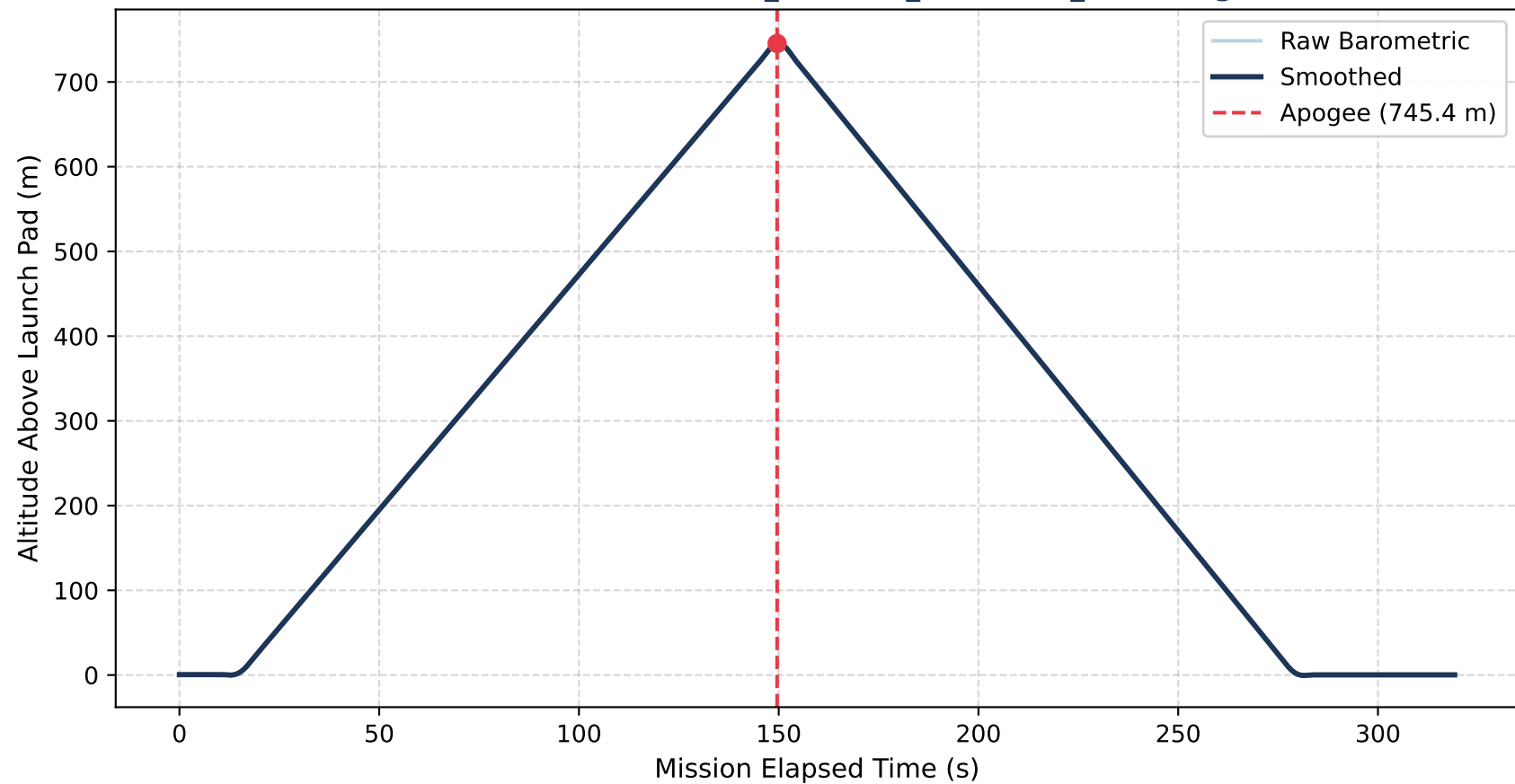


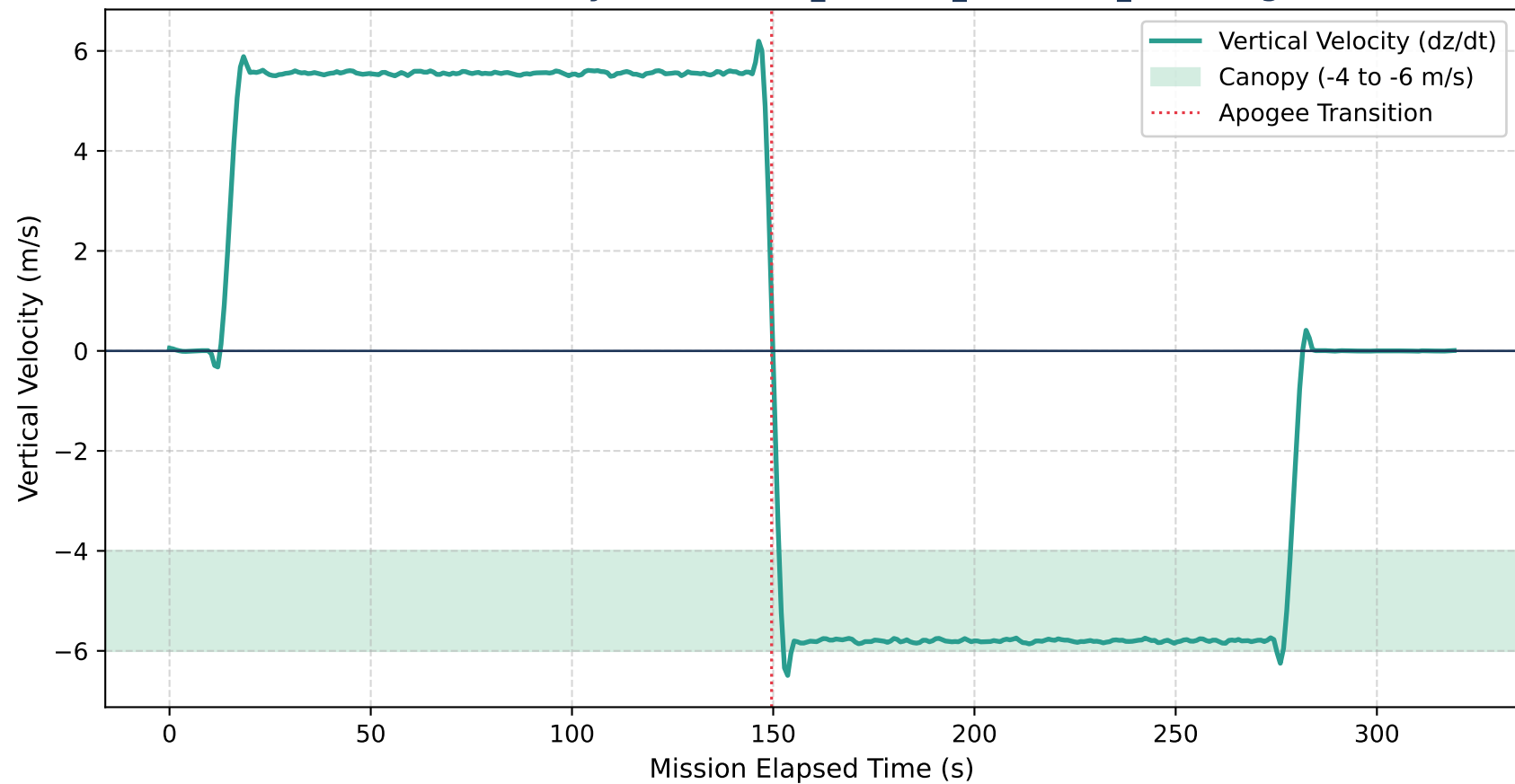
Cognitive CanSat - Mission Summary: 07_thermal_inversion_sounding

Flight Dynamics Parameter	Calibrated Value
Mission Name	07_thermal_inversion_sounding
Apogee (m)	745.44
Time to Apogee (s)	149.6
Max Ascent Velocity (m/s)	6.19
Max Descent Velocity (m/s)	-6.49
Max G-Shock (Gs)	1.01
Touchdown Shock (Gs)	1.01
Mission Duration (s)	319.2
Min Temperature (°C)	20.0
Max Temperature (°C)	24.52
Min Pressure (hPa)	926.46
Max Pressure (hPa)	1013.32
Lapse Rate (°C/100m)	0.606
Telemetry Frames	400
Landing Lat	22.575674
Landing Lon	88.369015

Altitude vs. Time — 07_thermal_inversion_sounding

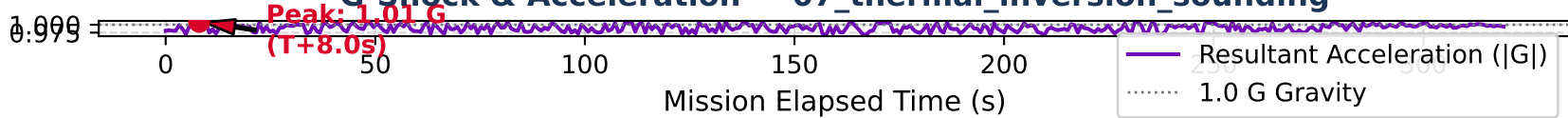


Vertical Velocity Profile — 07_thermal_inversion_sounding



Acceleration Magnitude (Gs)

G-Shock & Acceleration — 07_thermal_inversion_sounding



Atmospheric Sounding (Lapse Rate: 0.606 °C/100m)

Ambient Temperature (°C)

20

21

22

23

24

Altitude Above Pad (m)

700
600
500
400
300
200
100
0

Pressure (hPa)
Temperature (°C)

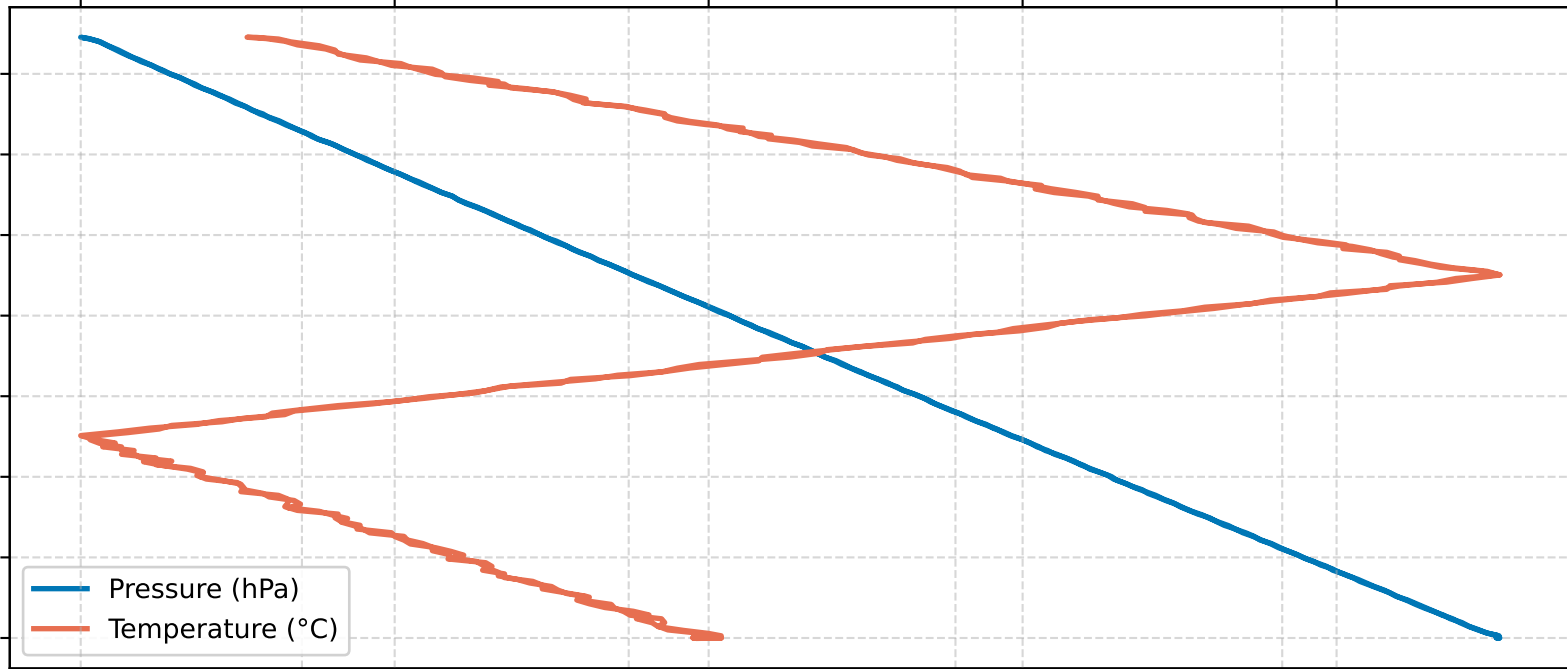
940

960

980

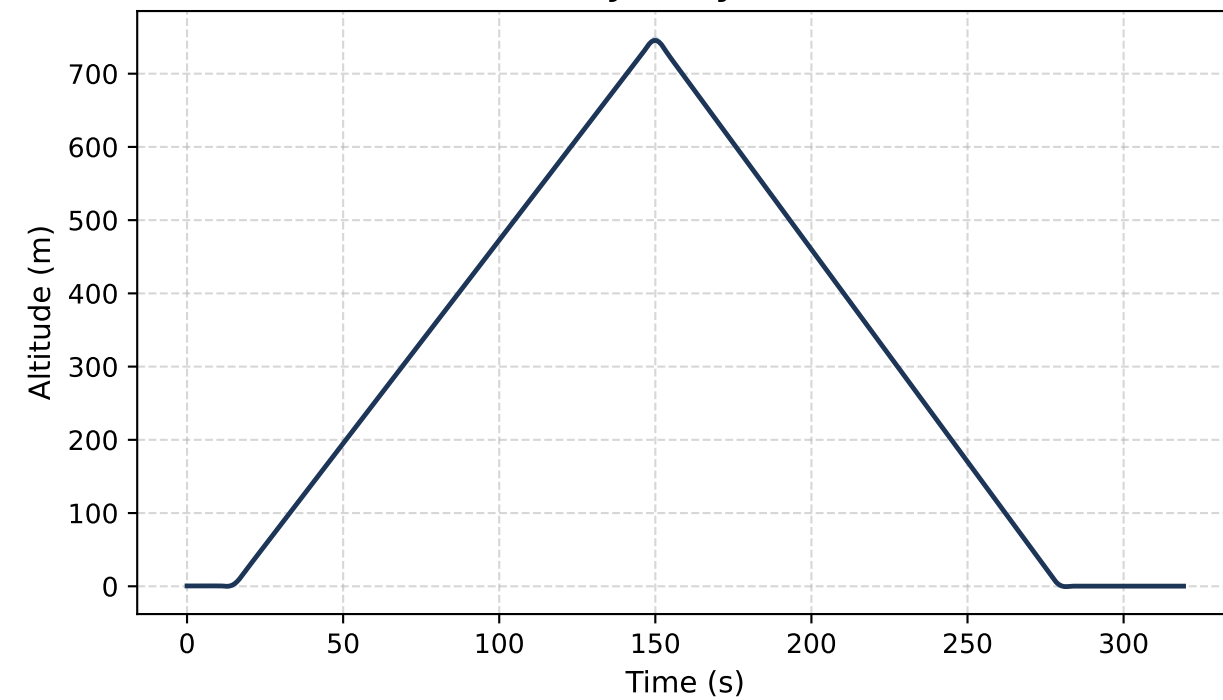
1000

Barometric Pressure (hPa)

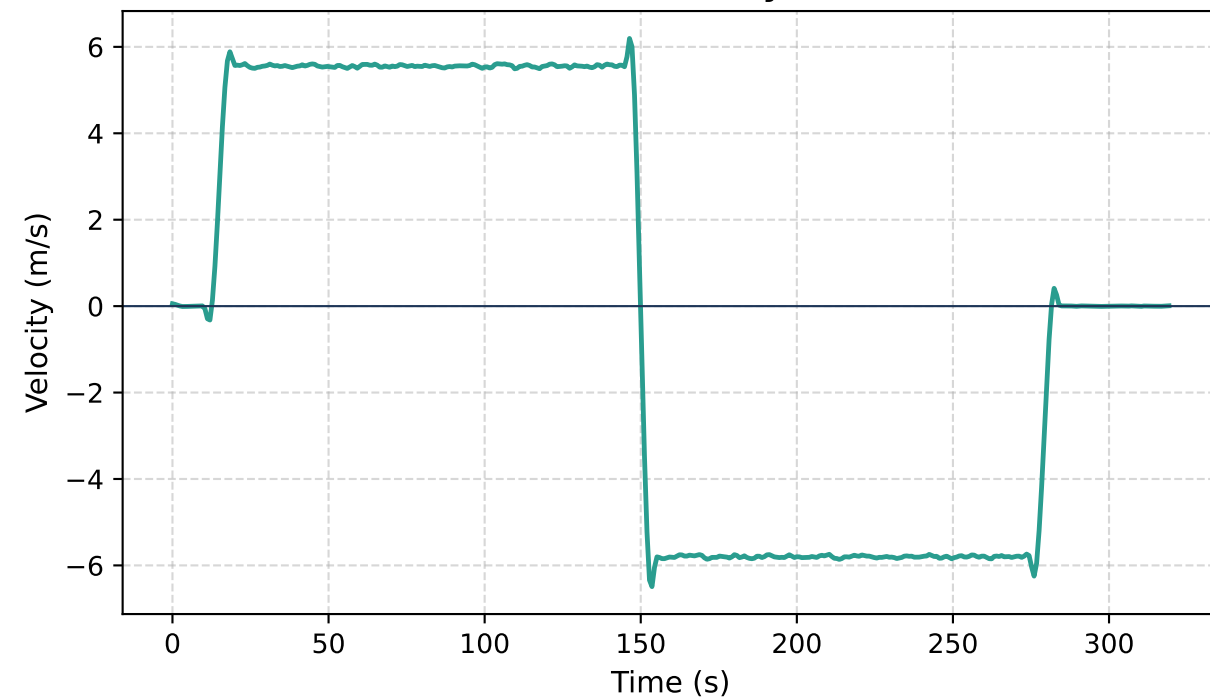


Cognitive CanSat Mission Dynamics: 07_thermal_inversion_sounding

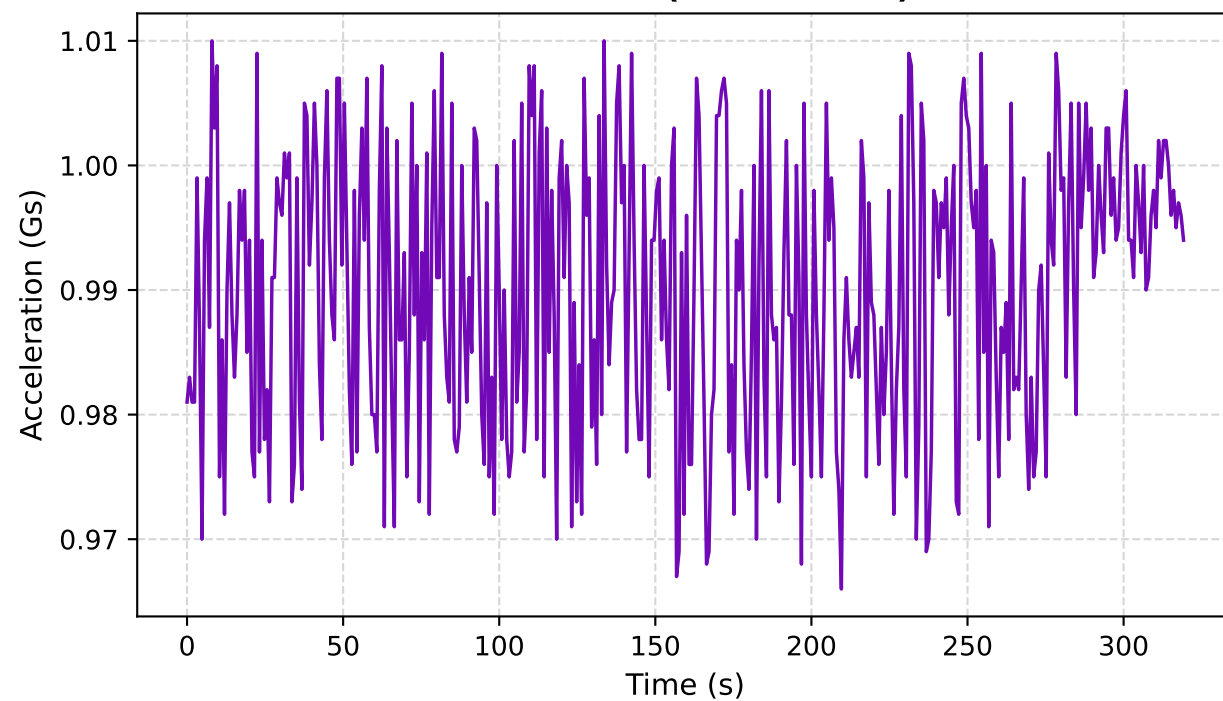
A. Trajectory Profile



B. Vertical Velocity Curve



C. G-Shock (Peak: 1.01G)



D. Atmospheric Sounding Profile

